

Quarterly Trading Patterns of Financial Institutions

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Abstract

This paper investigates whether different types of institutions have discernible trading motives in response to portfolio disclosures. Results show that banks, life insurance companies, mutual funds, and investment advisors who act as external managers generally have similar trading strategies. They sell more poorly performing stocks during the fourth quarter than the first three quarters of the year, and such trading behavior is more pronounced for institutions whose stocks on average have underperformed the market. In contrast, property and liability insurance companies, internally-managed pension funds, colleges, universities, and foundations, who mainly provide their own asset management services, show less inclination to window dress their portfolios.

I. Introduction

The recent growing institutional dominance of the equity markets in the United States has attracted considerable attentions from the media and many academics.¹ Of particular interest is the information on portfolio holdings by institutions disclosed under Section 13(f) of the Securities Exchange Act of 1934 — institutional investors with investment discretion over \$100 million in equity securities are required to report their holdings to the Securities and Exchange Commission (SEC) on a quarterly basis. Conventional wisdom is that 13(f) institutional investors have incentives to “window dress” their portfolios. Window dressing is a form of portfolio manipulation in which money managers routinely buy and sell securities in advance of public disclosure dates to “hide” their mistakes or to highlight their seemingly astute picks.² This could be done by dumping losing stocks, or loading up on winners, or both.³

Lakonishok, Shleifer, Thaler, and Vishny (1991) find that the portfolio strategies of 769 pension funds are consistent with the window-dressing strategy. Haugen and Lakonishok (1988) show that window dressing by institutional investors possibly accounts for the relatively high average stock returns displayed in late December and in early January (i.e., the turn-of-the-year effect). Musto (1999) provide evidence that money market funds hold disproportionately more government securities just before, and disproportionately fewer just after, the dates on which the funds publicly disclose their portfolios in order to make them appear safer and more conservative than they really are. Chevalier and Ellison (1997) examine portfolio holdings of growth and growth and income funds and find that the funds alter the composition of their portfolios at the end of the year towards high-quality, less-risky stocks. Supposedly, such a strategy will help the funds to attract more

¹Gompers and Metrick (2001) show that the equity holdings of institutional investors have substantially increased from \$375 billion in 1980 to \$3.98 trillion in 1996. Their figures indicate that large institutions held about 51% of the equity market share in the United States in 1996, as compared to 27.6% in 1980.

²Throughout the paper, the terms “institutional investors”, “institutional managers”, and “institutions” are used synonymously to refer to 13(f) institutions or managers.

³See a Morningstar.com’s article by Peter Di Teresa, “Is your fund’s portfolio just for show?”, January 4, 2002.

new investments and hence increase managers' assets under management. If institutions window dress their portfolios in anticipation of quarterly disclosures, then the holdings they report might reveal some unusual trading behavior, especially during the last quarter of the year.

This paper investigates whether institutions respond strategically to portfolio disclosure requirements and whether different types of institutions exhibit similar trading strategies. In particular, we look for evidence of window dressing activity by analyzing the equity holdings that institutions must disclose at the end of each calendar quarter, especially at the end of the year. Institutions such as banks, life insurance companies, mutual funds, and investment advisors have a fiduciary duty to prudently invest funds on behalf of their clients. Others such as private and public pension funds, property and liability insurance companies, colleges, universities, and foundations manage primarily their own assets. O'Barr and Conley (1992) point out that banks, insurance companies, and investment advisors and companies are more inclined to emphasize short-term performance in order to compete with one another for new clients (and, hence, more new money into the funds) than internally-managed pension funds, colleges, university endowments, and foundations. Their evidence might reflect the agency relationship between institutions and their clients. The former institutions generally act in their own interest and have strong incentives to strategically window dress their portfolios in order to attract new clients and hence increase new investments. On the other hand, the latter, who make their own investments, would be more concerned about performance and thus less inclined to engage in window dressing, which may prove costly in terms of transaction costs. In contrast to many existing studies, we specifically investigate whether the roles of the two types of money managers have any effect on their trading strategies in response to mandatory quarterly portfolio disclosures. To facilitate our discussion throughout the paper, we label the institutions who manage funds on behalf of their customers as 'external money managers' or 'external managers' and those who self manage their own money as 'internal money managers' or 'internal managers'.

Our results show that in every quarter, institutional managers sell more of both ex-

treme loser and winner stocks, but sales of extreme loser stocks are larger than sales of extreme winner stocks. In particular, institutions whose portfolios have underperformed the market tend to sell more poorly performing stocks in proportion to their holdings than those whose portfolios have outperformed the market. Sales of such loser stocks by ‘loser’ managers rise significantly from 66 percent in quarters 1-3 to 77 percent in quarter 4, as compared to 17 percent and 13 percent by ‘winner’ managers. This evidence is consistent with the window-dressing hypothesis. The results also show that managers who manage funds on behalf of their clients are likely to sell more extreme loser stocks but fewer extreme winner stocks during the fourth quarter of the year, a strategy consistent with window dressing. Such trading behavior is more pronounced for managers whose stock portfolios have underperformed the market than for those whose stock portfolios are not. By contrast, there is weaker evidence that internal managers exhibit such trading behavior. Furthermore, we find that both external and internal managers whose portfolios have outperformed the market are typically contrarians (buy losers and sell winners). Overall, the patterns of institutional equity holdings reveal interesting differential trading motives between winner and loser managers and between externally- and internally-managed funds.

The paper is organized as follows. Section II describes the data. In Section III we test the window-dressing hypothesis by examining institutional sales and purchases of extreme loser and winner stocks, as revealed by reported quarterly stock holdings. The final section concludes.

II. Data

The institutional disclosure program mandated by Section 13(f) of the Securities Exchange Act of 1934 requires large institutional investment managers to file information about equity holdings under their investment discretion to the SEC. In 1978 the SEC adopted rule 13(f-1) and form 13F. The rule requires institutional managers exercising investment discretion over \$100 million in equity securities to file the information prescribed

by form 13F with the SEC. The originally proposed rule required the form to be filed annually, but since 1979 the final version of the rule has required quarterly filings. The reporting threshold encompasses various types of institutional managers such as banks, investment companies, pension funds, insurance companies, and brokerage houses. The reports contain information about equity holdings of reporting managers and the type of investment discretion they and their other investment managers exercise with respect to those securities. These reports are available from CDA Investment Technologies, Inc., a service company engaged by the SEC to process and maintain 13(f) filings.

This study uses CDA's quarterly institutional holdings data from March of 1986 to September of 1998. The data set also includes the net sales and purchases of stocks that an institution makes during a given quarter. As such, we do not have information about the number of times an institution might have bought and sold the same stock within that quarter. Lakonishok et al. (1991), however, have documented that the turnover ratio calculated using the quarterly change of stockholdings is typical for institutions, thus suggesting that net transactions provide a reasonably good proxy for institutional sales and purchases. Furthermore, the data contain no additional information with respect to the manner in which they categorize the institutions, and hence our analysis shall have to rely on their classifications as a means to differentiate managers' trading motives.

In the 13(f) institutional filings database, the classifications of institutional managers are (i) banks, (ii) insurance companies, (iii) investment companies (mutual funds), (iv) investment advisors (large brokerage houses), and (v) others, which include pension funds, colleges, university endowments, and foundations. Banks manage funds on behalf of private trust and pension plan clients. They are most affected by the common-law prudent-man rule that induces them to invest in stocks that courts would easily consider as prudent (Del Guercio (1996)). Insurance companies are mainly of two types: Life insurance companies predominantly manage their clients' private pension funds, and property and liability insurance companies essentially manage their own funds. Insurance managers, on the other hand, are not only constrained by the prudent-man rule, but also restricted from investing more than a specified fraction of their managed assets in equities. But

they are subject to less stringent fiduciary constraints (Badrinath, Gay, and Kale (1989)). Investment companies and advisors manage individual assets through mutual funds and also act as external money managers for pensions and endowments. CDA classifies an institution as a mutual fund if the money it manages is more than 50 percent in mutual fund assets. Otherwise, CDA categorizes it as the investment advisor type. Investors find investing in these two types of institutions more attractive, because withdrawal of funds generally can be made on demand.

Banks, life insurance companies, investment advisors, and investment companies mainly manage assets on behalf of their clients. Such a manager-customer relationship inevitably causes a potential conflict of interests between them – a typical agency problem. Actions taken by these institutions to maximize their own interests generally might be inconsistent with those they should implement in order to maximize the interests of their clients. The existence of agency costs possibly sets apart the trading motives of these institutions from the trading motives of institutions such as private and public pensions, colleges, property and liability insurance companies, and endowments of both foundations and universities, whose assets are mostly managed internally.

We limit our study to institutional holdings of all common stocks traded on the NYSE, AMEX, and NASDAQ and returns data of these stocks which are available from the Center for Research in Security Prices (CRSP). The sample period is from January 1986 through December 1998. We also focus on those timely 13(f) filings as well as all late-filings. Under Rule 13f-1(a), any filing that is not made within 45 days after the last day of each quarter-end is considered a late filing. Late filings are granted under special circumstances that the SEC views necessary in the public interest or for the protection of investors.⁴ For example, a late filing can be requested if an institution is engaged in an ongoing investment strategy that involves a large position change on the reporting date.

Table 1 presents descriptive statistics on the 13(f) institutional filing database, which

⁴We thank the referee for this point. For more information, see “Commission Notice: Re: Section 13(f) Confidential Treatment Requests,” dated June 17, 1998; the article is available from <http://www.sec.gov/divisions/investment/guidance/13fpt2.htm>.

is broken down by type of institution. Panel A contains the number of institutions that report their portfolio holdings to the SEC at the end of 1986 through 1997. Over this 12-year period, investment advisors have the largest number of qualifying institutions. The number increases from 381 in 1986 to 1,047 in 1997. The investment advisors group is about 26 percent and 71 percent of the total number of institutions in 1986 and 1997, respectively. Banks have the second largest number, but the number has been declining from 209 in 1986 to 180 in 1997. The banks' proportion of the total number of institutions decreased from 26 percent in 1986 to 12 percent in 1997. Each of the remaining three manager classifications, however, represents about 10 percent or less of the total number of 13(f) institutions. Overall, the total number of institutions has grown from 818 in 1986 to 1479 in 1997. This increase is in part due to the rising stock market value, which results in many smaller institutional investors hitting the minimum requirement of the \$100 million in holdings and having to report their holdings to the SEC.

Panel B shows the percentage of the total market value of institutional holdings in each equity market and the dollar amount of institutional holdings by manager type. In 1986, banks and investment advisors together have the largest dollar amount of institutional holdings: \$373 and \$396 billion, respectively, together representing 72 percent of the total value of institutional holdings. However, in 1997, investment advisors and mutual funds have the largest share; their dollar amounts are about \$1.5 trillion and \$2 trillion, together representing 60 percent of the total market value of institutional holdings. Regardless of the size of total investments, institutions typically have at least 80 percent of their holdings in NYSE stocks, about 16 percent in NASDAQ stocks, and less than two percent in AMEX.

Table 2 shows for each type of institution the percentage of the average market value of institutional holdings in NYSE, AMEX, and NASDAQ stocks across five size-formed quintiles.⁵ The table reveals some distinct patterns in the institutional holdings across

⁵We used NYSE stocks to determine the breakpoints for the five size quintiles and then assigned all the NYSE, AMEX, and NASDAQ stocks into each size-formed quintile according to their market capitalization.

size-formed quintiles and across manager types. For their holdings in NYSE stocks, all institutions have the highest percentage concentrated in the largest size-formed quintile and the lowest percentage in the smallest size-formed quintile. Banks and the “others” category of institutions (public and private pensions, colleges, universities, and foundations) hold about 86 to 87 percent of the largest NYSE stocks, while insurance companies, mutual funds, and investment advisors hold about 78 to 80 percent. In contrast, for the AMEX and NASDAQ stocks, they hold about 20 percent in the largest AMEX stocks and about 37 to 52 percent in the largest NASDAQ stocks. Given more than 80 percent of their funds are invested in NYSE stocks (see Panel B of Table 1), the pattern suggests that institutions generally prefer large, high quality NYSE stocks, which are considered relatively less risky.

III. Trading Activity of Institutions

This section compares trading activity of five different types of institutions and across all quarters. Our approach contrasts with those of existing studies that have focused their analysis largely on institutional investors as a group. These disparate institutions, however, have significantly different investment incentives that are likely to influence their trading motives. Banks, investment advisors, mutual funds managers, and life insurance companies face the need to stay competitive relative to other managers by way of investing in typically more risky stocks in order to generate greater expected returns, hence maximizing their expected performance-based remuneration on the one hand, and the desire to conceal their risky strategies at the time when reporting their stock holdings to the SEC and their clients, on the other. These money managers who manage assets mostly on behalf of their customers are more concerned about their own interests and have strong incentives to actively window dress their portfolios at quarter-end disclosures. However, property and liability insurance companies, colleges, universities, foundations, and public and private pension funds who render their own asset management services obviously have less of a need to rebalance (or window dress) their portfolios at quarter-ends. By

examining the quarterly institutional stock holdings, we can determine whether there are any differences in the trading motives between the two groups of money managers.

A. Measures of Trading Activity

Our analysis applies Lakonishok et al.'s (1991) approach to measure trading activity by institution. In order to implement this approach, we must first classify stocks based on their return performance. At the end of each quarter, we employ NYSE stocks to determine the breakpoints for five size-based portfolios and then assign all NYSE, AMEX, and NASDAQ stocks into each size-based quintile according to their market capitalization. Within each size-based quintile, we further divide the stocks into five equal groups based on stock returns over the past year up to the end of that quarter. Then we re-group the resulting 25 portfolios into five quintiles based on stock performance. This is done by gathering the same performance stocks from each size-based portfolio together to form a group. For example, the lowest performance-based portfolio contains the lowest 20 percent performing stocks from each size-based portfolio, and the highest performance portfolio contains the best 20 percent performing stocks from each size-based portfolio. As a result, each performance quintile will have the same distribution of market capitalization. This procedure is repeated every quarter. Throughout this study, these five performance quintiles are employed as the base portfolios.

Given that we know the stocks within each performance quintile, we can compute how much each manager holds, buys, and sells every quarter from each quintile. The sale of stocks in performance quintile j by each institution k during a quarter q is measured by

$$\frac{SELL(j, q, k)/HOLD(j, q - 1, k)}{\sum_j SELL(j, q, k)/\sum_j HOLD(j, q - 1, k)}, \quad (1)$$

where $SELL(j, q, k)$ is the dollar value of sales by institution k in quarter q and in performance quintile j , and $HOLD(j, q - 1, k)$ is the dollar value of holdings at the end of the previous quarter of the same stocks as those in performance quintile j in quarter q . Values of $SELL$ and $HOLD$ are calculated as the average of the beginning and end of quarter q prices. (1) shows the ratio of sales in a performance quintile to holdings of

the same stocks at the end of the previous quarter divided by the ratio of total sales in quarter q to holdings at the end of the previous quarter. The ratio given by (1) measures the selling intensity of an institution in a given performance quintile relative to the overall selling intensity.

Similarly, the purchase of stocks in performance quintile j by each institution k during a quarter q is measured by

$$\frac{BUY(j, q, k) / \sum_j BUY(j, q, k)}{UNIV.HOLD(j, q) / \sum_j UNIV.HOLD(j, q)}, \quad (2)$$

where $BUY(j, q, k)$ is the dollar value of purchases by institution k in quarter q and in performance quintile j , and $UNIV.HOLD(j, q)$ is the value of CRSP universe holdings in quarter q and in performance quintile j . Values of BUY and $UNIV.HOLD$ are also computed as the average q prices as of the beginning and end of the quarter. (2) shows the fraction of an institution's purchases in a performance quintile relative to the fraction of the universe holdings in the same performance quintile.

For every quarter, we calculate (1) and (2) for each institution, weight them by the size of its holdings relative to the total size of all institutional holdings, and then sum them across all institutions. Finally, we average the two variables over quarters 1-3 and over quarter 4, and next compare the average over quarters 1-3 to that for quarter 4.

B. Activity of Institutions as an Aggregate

In this section, we examine whether institutional managers in aggregate engage in window dressing at disclosures. Table 3 presents value-weighted average sales and value-weighted average purchases of stocks within each performance quintile that all institutions make during quarters 1-3 and quarter 4. The table also shows t -tests for the difference in the means of the trading measures between quarters 1-3 and 4. Standard errors of the trading measures are reflected in parentheses.

We first look at the findings for quarters 1-3. Institutions, on average, buy 5.7 percent more of extreme winner stocks relative to the universe of available stocks (ratio is 1.057)

and 24.3 percent more of extreme loser stocks (ratio is 1.243). During the same period, they also sell 32.2 percent more of extreme winners relative to their holdings (ratio is 1.322) and 38.5 percent more of extreme losers (ratio is 1.385). The findings of large purchases of extreme losers and large sales of extreme winners probably suggest that institutions on average are contrarians. On the other hand, large institutional sales of extreme loser stocks provide evidence that institutions do window-dress their portfolios for reporting purposes. In contrast, there is relatively much less buying and selling of stocks in the middle three performance quintiles.

We then compare the findings in quarters 1-3 with the findings in quarter 4. Institutions sell about 5 percent more of extreme loser stocks, but 9 percent less of extreme winner stocks in quarter 4 than in quarters 1-3. They also buy 2 percent more of extreme loser stocks, but there is no statistical difference in the purchase of extreme winner stocks. The unusually large institutional sale of extreme loser stocks, especially in the fourth quarter, is consistent with the window-dressing hypothesis.

On average, the difference in aggregate trading activity between quarters 1-3 and 4, while statistically significant, is small. The results, however, have not considered the effects of institutional past performance on their holdings at disclosures. It is therefore of interest to examine whether the institutions' prior return performance will have any bearing on their trading strategies. To an analysis of this we next turn.

C. Activity of Winner and Loser Institutions

In this section we specifically examine the trading behavior of winner and loser institutions. We hypothesize that institutions whose stocks on average have underperformed a benchmark index ('loser' institutions) would become more aggressive with their holdings around disclosure dates and are more likely to window dress their portfolios than those whose stocks on average have outperformed the benchmark index ('winner' institutions). Hence one would expect window dressing to be more evident in loser than winner institutions. To test whether past performance affects institutional investors' trading strategies

around portfolio disclosure dates, we sort the institutions into four groups based on their return performance relative to the return of a market index. Winner (loser) institutions represent the top (bottom) quartile of institutions whose accumulated stock returns from the beginning of a calendar year up to the end of that quarter is greater (less) than the return on the CRSP market index. Table 4 reports the value-weighted average sales of stocks in two extreme performance quintiles that winner and loser institutions make relative to their own holdings during quarters 1-3 and quarter 4 and value-weighted average purchases of extreme loser and winner stocks relative to their universe holdings. It also shows the t -test for the difference in the means of trading measures of winner and loser institutions between quarters 1-3 and 4.

There emerge a few interesting differences between winner and loser institutions. During quarters 1-3, winner institutions sell 17 percent more of extreme loser stocks and 46 percent more of extreme winner stocks. Correspondingly, however, loser institutions sell 66 percent more of extreme loser stocks and 2 percent less of winner stocks. The results suggest that loser institutions have a greater tendency to sell more loser stocks than winner institutions. On the other hand, winner institutions tend to sell more winner stocks than loser institutions. Thus, as compared to their winner counterparts, loser institutions engage more actively in window dressing their portfolios for quarter-end reporting purposes. For the same period, winner institutions buy about 30 percent more of loser stocks and only about 9 percent more of winner stocks. In contrast, however, loser institutions buy about 22 percent more of loser stocks and about 4 percent less of winner stocks in quarters 1-3. The findings from the first three quarters also suggest traces of evidence that winner institutions in general engage in contrarian strategies; they buy more extreme loser stocks and sell more extreme winner stocks.

We next compare trading activity in quarter 4 to trading activity in the previous three quarters. Sales of loser stocks by winner institutions fall by about 5 percent from quarters 1-3 to quarter 4, whereas sales of the same by their loser counterparts rise by about 11 percent. Specifically, loser institutions sell 77 percent more of loser stocks during the fourth quarter as compared to 66 percent more in quarters 1-3. In comparison,

winner institutions sell 13 percent more loser stocks in quarter 4 vis-à-vis 17 percent more in quarters 1-3. Intuitively, institutions that have performed poorly relative to the market in the past quarter are more likely to rebalance their portfolios by getting rid of poorly performing stocks than institutions that have performed well relative to the market. Hence, the results show that loser institutions account for a substantial part of the sales of extreme loser stocks during the last quarter of the year, suggesting that they window dress their portfolios more than winner institutions.

Further, the table shows no statistical difference in the sales and purchases of winner stocks by winner institutions between quarters 1-3 and 4. On the contrary, loser institutions sell about 10 percent less of winner stocks during the fourth quarter than the first three quarters of the year and buy about 2 percent less of winner stocks as well. The results suggest that loser institutions are less inclined to buy more of the winner stocks. We interpret this finding as evidence that winner stocks are probably overvalued and hence loser institutions find these stocks to be too expensive.

D. Activity of Internal and External Money Managers

So far, the evidence on institutional sales and purchases of winner and loser stocks is broadly consistent with the window-dressing hypothesis. However, the findings are based on trading by institutions as a group, without differentiating the type of institution. As we discussed earlier, banks, life insurance companies, mutual funds, and investment advisors, who primarily manage assets on behalf of their clients, tend to focus more on short-term performance in order to remain competitive. On the other hand, there is no reason for property and liability insurance companies, university endowments, colleges, foundations, and pension funds, who essentially manage their own funds internally, to do the same. Conceivably, these two groups of managers might display different trading behavior. This section addresses the issue.

Table 5 presents results on trading activity of external and internal money managers, with standard errors in parentheses. It also reports t -tests of equality of means between

quarters 1-3 and quarter 4. Similar to Table 4, this table summarizes the respective value-weighted average sales and purchases of extreme loser and winner stocks by manager type and by quarter. Each group of institutions is further divided into winner and loser managers. The table shows some distinctive trading strategies by external and internal money managers.

One, external money managers sell about 44 percent more of extreme loser stocks in quarters 1-3 and 57 percent more in the last quarter; the difference of 13 percent is found to be statistically significant at the five percent level. The active selling of loser stocks by external managers is mainly attributable to the sales made by those whose portfolios have underperformed the market index in the past year. In particular, loser external managers oversell loser stocks by 71 percent in quarters 1-3 and 86 percent in quarter 4, while winner external managers oversell loser stocks by 26 percent in the first three quarters and by 15 percent in the last quarter. Loser managers increase, whereas winner managers decrease, the sales of loser stocks from quarters 1-3 to quarter 4. Thus loser managers are more aggressive in getting rid of loser stocks than managers who have performed well relative to the market. The finding is consistent with the hypothesis that loser institutions ought to sell more loser stocks if they do not wish to show that they have made too many “bad decisions”.

Unlike external money managers, all internal money managers sell only 24 percent more extreme loser stocks in the first three quarters and 27 percent more in the fourth quarter; the difference in the sales measures across quarters is statistically insignificant. The sales of these loser stocks are mainly attributable to the relatively active selling of such stocks by loser internal managers. The latter sell 43 percent more of loser stocks in quarters 1-3 and 47 percent more in quarter 4, as compared to 9 percent and 7 percent by winner internal managers. Thus, winner internal managers reduce sales of loser stocks by 2 percent across the quarters, whereas loser internal managers increase sales of the same by 4 percent.

The results therefore show contrasting differences between internal and external man-

agers. Selling of loser stocks by both winner and loser internal managers is substantially lower than that by winner and loser external managers. For both types of winner managers, there seems little evidence that they window dress their portfolios at quarter-ends. On the other hand, loser external managers sell significantly more loser stocks than loser internal managers; the differences of 28 percent (quarters 1-3) and 39 percent (quarter 4) in their sales ratios suggest that the former display strong evidence of window dressing activity.

Two, there is a small increase in the purchase of extreme loser stocks by both external and internal managers across quarters. Loser external managers not only sell, but also buy more loser stocks. In quarters 1-3, they buy 23 percent more of loser stocks and in quarter 4, this ratio posts a small rise of 2 percent. A similar pattern is observed in the trading behavior of winner external managers. They buy 20 percent more of loser stocks during quarters 1-3 and 27 percent more in quarter 4. In contrast to their counterparts, loser internal managers do not increase their purchases of loser stocks across the quarters, but winner internal managers increase theirs from 32 percent in quarters 1-3 to 37 percent in quarter 4. Increasing the purchases of loser stocks during quarter 4 might be an indication that such stocks are now ‘good buys’.

Finally, there is typically much less selling and buying of winner stocks by both types of loser managers. Comparing winner external and internal managers, we observe that winner external managers tend to buy and sell substantially more of extreme winner stocks than winner internal managers. The former sell 52 percent in quarters 1-3 and 56 percent in quarter 4, as compared to 23 percent and 19 percent by winner internal managers. Unlike winner managers, both types of loser managers sell and buy less of winner stocks. This evidence therefore indicates that selling losers to get rid of “mistakes” is a more popular means of window dressing than buying winner stocks to show their astute picks.

In summary, external managers, especially those whose portfolios have performed poorly relative to the market, sell poorly performing stocks disproportionately to their holdings. Such sales are observed to increase significantly in the last quarter of the year,

consistent with the window-dressing strategy. On the contrary, loser internal managers engage in a small increase of sales of loser stocks across the quarters, a strategy which is less consistent with window dressing. Interestingly, purchases of loser stocks by managers whose portfolios have performed better than the market also increase in the fourth quarter, an evidence suggesting that the managers are either taking advantage of the “bargain” loser stocks by loading more of them in their portfolios, or providing liquidity to individual investors who are selling their loser stocks. At the same time, they also sell more winner stocks. The strategy of buying more extreme losers and selling more extreme winners implies that external and internal managers whose portfolios have outperformed the market are typically contrarians. In contrast, there is much less selling and buying of winner stocks by loser managers, regardless of type.

IV. Conclusion

Existing studies have generally examined institutional investors as a group and considered their trading activity as a concerted effort. This study takes a different approach by testing whether institutional managing of assets, internally or externally, has any bearing on trading activity and investment decisions in anticipation of quarterly portfolio disclosures. Results indicate that on average some institutional investment decisions are affected by the portfolio disclosure requirement. We show that institutions who invest assets largely on behalf of their clients have a greater tendency to window dress their portfolios at disclosures than do institutions whose assets are primarily managed internally. We also show that this result is stronger for institutions whose stocks on average have underperformed the market. The results also suggest that institutions whose stocks on average have outperformed the market do engage in contrarian strategies.

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Table 1**Descriptive Statistics of the Institutional Holdings Database**

The table presents descriptive statistics of the institutional holdings database from March 1986 to December 1997. Panel A presents the number of institutions reporting their portfolio holdings by year, and Panel B presents the percentage of the total market value of institutional holdings in NYSE, AMEX, and NASDAQ stocks and also the total market value in billions of dollars at the end of December of each year by manager type and by exchange. Manager types include banks, insurance companies (life insurance and property and liability insurance companies), mutual funds, investment advisors, and other institutions (university and college endowments, foundations, and pensions).

Panel A: Number of Institutions By Year												
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Banks	209	209	214	217	215	213	216	192	194	202	172	180
Insurance												
Life Insurance	46	51	43	48	49	45	48	43	51	52	46	48
Property & Liab.	20	23	20	22	25	26	23	22	29	28	25	29
Mutual Funds	66	60	60	54	57	56	63	63	54	96	90	87
Investment Advisors	381	441	455	506	543	588	667	659	746	854	916	1047
Others	96	99	92	93	92	83	83	72	72	80	74	88
Total	818	883	884	940	981	1011	1100	1051	1146	1312	1323	1479

Table 1 (continued)

Panel B: % of the Market Value of Stockholdings by Type of Institution												
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Banks												
% in NYSE	94.6	94.3	93.9	93.9	95.0	93.9	93.3	92.0	91.9	91.4	89.5	89.6
% in AMEX	0.6	0.7	0.7	0.7	0.5	0.5	0.3	0.4	0.4	0.3	0.3	0.2
% in NASDAQ	4.7	5.1	5.4	5.4	4.4	5.6	6.3	7.6	7.7	8.3	10.2	10.1
Market Value (\$b)	372.5	344.2	374.8	456.0	396.2	534.9	558.3	626.7	598.3	751.9	906.6	1191.1
Insurance												
% in NYSE	94.1	92.6	93.0	92.7	93.1	90.2	90.5	89.0	89.8	87.5	85.4	86.7
% in AMEX	1.0	0.9	0.9	1.2	1.0	1.4	0.5	0.6	0.6	0.6	0.5	0.4
% in NASDAQ	5.0	6.5	6.1	6.0	5.9	8.4	9.0	10.4	9.6	11.9	14.1	12.9
Market Value (\$b)	89.1	83.9	95.8	115.3	105.3	138.1	152.9	205.0	224.0	328.9	397.4	547.5
Mutual funds												
% in NYSE	91.1	89.8	90.0	89.8	89.8	86.4	86.8	85.9	84.5	83.2	83.3	85.9
% in AMEX	1.3	1.3	1.3	1.6	1.3	1.3	0.8	1.1	0.7	0.7	0.6	0.5
% in NASDAQ	7.6	8.9	8.7	8.6	8.9	12.3	12.4	13.0	14.7	16.1	16.1	13.6
Market Value (\$b)	66.6	64.2	70.2	84.4	84.7	174.2	220.8	287.0	319.5	733.3	1022.7	1471.3
Investment adv.												
% in NYSE	92.2	91.5	91.2	90.6	91.0	87.9	86.7	85.5	84.9	84.1	82.9	84.3
% in AMEX	1.3	1.2	1.1	1.2	0.9	1.0	0.7	1.0	0.9	0.8	0.7	0.7
% in NASDAQ	6.5	7.3	7.7	8.3	8.1	11.1	12.5	13.6	14.3	15.0	16.4	14.9
Market Value (\$b)	395.6	407.2	467.7	588.4	585.9	777.4	898.9	1036.6	1019.4	1204.5	1548.9	1978.9
Others												
% in NYSE	96.9	96.0	94.5	95.4	95.8	94.5	93.8	92.8	91.8	90.3	88.4	89.3
% in AMEX	0.6	0.8	1.1	1.0	0.9	0.9	0.5	0.7	0.6	0.5	0.3	0.3
% in NASDAQ	2.4	3.2	4.4	3.6	3.3	4.6	5.7	6.5	7.5	9.3	11.2	10.4
Market Value (\$b)	136.7	141.3	165.4	204.9	180.5	230.2	238.8	247.1	256.4	302.2	331.0	437.3

Table 2

**Percent of the Average Market Value of Quarterly Institutional Holdings in
NYSE, AMEX, and NASDAQ Stocks by Type of Institution**

At the end of each quarter, all stocks traded on NYSE, AMEX, and NASDAQ are sorted into size quintiles according to five NYSE breakpoints. The table shows the percent of the average market value of each group's quarterly stockholdings in NYSE, AMEX, and NASDAQ stocks across five size-formed quintiles. The entire sample period is from March 1986 through December 1997.

	ALL	NYSE	AMEX	NASDAQ
Smallest				
Banks	0.81	0.21	14.23	6.89
Insurance	0.96	0.27	14.19	5.16
Mutual Funds	1.42	0.39	6.79	6.69
Investment Advisors	2.04	0.57	14.33	10.30
Others	0.67	0.16	10.47	5.79
Quintile 2				
Banks	1.89	1.02	20.75	11.65
Insurance	2.52	1.48	14.29	10.55
Mutual Funds	3.16	1.60	16.90	11.65
Investment Advisors	3.88	2.09	17.14	14.90
Others	1.69	0.88	11.16	11.96
Quintile 3				
Banks	4.03	2.93	20.34	16.77
Insurance	5.74	4.59	15.49	14.88
Mutual Funds	6.73	4.93	21.84	16.54
Investment Advisors	7.03	5.28	18.84	17.84
Others	3.62	2.62	25.35	15.17
Quintile 4				
Banks	10.94	10.03	23.80	21.45
Insurance	14.41	13.85	36.13	17.60
Mutual Funds	15.57	14.42	32.50	21.46
Investment Advisors	14.83	13.88	29.71	20.04
Others	10.38	9.45	32.11	21.09
Largest				
Banks	82.33	85.81	20.88	43.24
Insurance	76.38	79.80	19.91	51.79
Mutual Funds	73.12	78.65	21.97	43.66
Investment Advisors	72.22	78.18	19.99	36.92
Others	83.63	86.89	20.92	45.99

Table 3**Institutional Sales and Purchases of Extreme Loser and Winner Stocks by Quarter**

The table shows value-weighted average of all institutional sales and purchases of stocks from five performance quintiles. The performance quintiles are constructed as follows. At each quarter, NYSE, AMEX, and NASDAQ stocks are sorted into equal quintiles based on stock return performance over the past year up to that quarter, with each quintile having the same distribution of market capitalization. ‘Sales’ are defined as the ratio of sales in a performance group to holdings of the same stocks at the end of the previous quarter divided by the ratio of total sales in this quarter to holdings at the end of the previous quarter. ‘Purchases’ are defined as the fraction of stocks an institution purchases in a performance group relative to the fraction of universe holdings in that performance group. Standard errors are reported in parentheses. *t*-test is for the test of equality of means between quarters 1-3 and quarter 4. The sample period is from March 1986 through December 1998.

	Sales			Purchases		
	Quarters 1-3	Quarter 4	<i>t</i> -Test	Quarters 1-3	Quarter 4	<i>t</i> -Test
Extreme Loser Stocks	1.385 (0.044)	1.430 (0.051)	0.045 (4.781)	1.243 (0.038)	1.265 (0.034)	0.022 (3.465)
Quintile 2	1.022 (0.020)	1.033 (0.030)	0.011 (1.565)	1.000 (0.015)	0.981 (0.025)	-0.019 (-2.777)
Quintile 3	0.945 (0.014)	0.974 (0.021)	0.029 (4.268)	0.913 (0.009)	0.917 (0.021)	0.004 (0.819)
Quintile 4	0.966 (0.017)	0.985 (0.023)	0.019 (2.263)	0.898 (0.013)	0.901 (0.020)	0.003 (0.532)
Extreme Winner Stocks	1.322 (0.060)	1.232 (0.053)	-0.090 (-7.52)	1.057 (0.024)	1.050 (0.028)	-0.007 (-1.194)

Table 4

Sales and Purchases of Extreme Loser and Winner Stocks by Quarter and by Winner and Loser Institutions

The table shows value-weighted average sales and purchases of stocks from two extreme performance quintiles by winner and loser institutions. The two extreme performance quintiles consist of extreme loser and winner stocks, respectively and are constructed as follows. At each quarter, NYSE, AMEX, and NASDAQ stocks are sorted into equal quintiles based on stock return performance over the past year up to that quarter, with each quintile having the same distribution of market capitalization. Winner (loser) institutions represent the top (bottom) quartile of institutions whose accumulated stock returns from the beginning of each calendar year to the end of that quarter in question is greater (less) than the return on the CRSP market index. Institutions are divided into four quartiles based on their returns relative to the CRSP market index. ‘Sales’ are defined as the ratio of sales in a performance group to holdings of the same stocks at the end of the previous quarter divided by the ratio of total sales in this quarter to holdings at the end of the previous quarter. ‘Purchases’ are defined as the fraction of stocks an institution purchases in a performance group relative to the fraction of universe holdings in that performance group. Standard errors are reported in parentheses. *t*-test is for the test of equality of means between quarters 1-3 and quarter 4. The sample period is from March 1986 through December 1998.

	Sales			Purchases		
	Quarters 1-3	Quarter 4	<i>t</i> -Test	Quarters 1-3	Quarter 4	<i>t</i> -Test
Extreme Loser Stocks						
Institutions						
Winners	1.171 (0.050)	1.126 (0.065)	-0.046 (-3.221)	1.301 (0.046)	1.360 (0.017)	0.058 (4.784)
Losers	1.657 (0.091)	1.766 (0.096)	0.109 (6.376)	1.215 (0.056)	1.223 (0.058)	0.008 (0.659)
Extreme Winner Stocks						
Institutions						
Winners	1.463 (0.063)	1.468 (0.077)	0.005 (0.245)	1.091 (0.068)	1.099 (0.062)	0.007 (0.384)
Losers	0.978 (0.037)	0.879 (0.058)	-0.099 (-9.053)	0.957 (0.014)	0.941 (0.020)	-0.016 (-1.906)

Table 5

Institutional Sales and Purchases of Extreme Loser and Winner Stocks by Manager Type

The table shows value-weighted average sales and purchases of stocks from two extreme performance quintiles (extreme loser and winner stocks, respectively) by external and internal money managers. Managers are further divided into winner and loser managers, as defined in Table 3. External money managers include banks, life insurance companies, mutual funds, and investment advisors, while internal money managers include property and liability insurance companies, colleges, universities endowments, foundations, and pension funds. Standard errors are indicated in parentheses. t -test is for the test of equality of means between quarters 1-3 and quarter 4. The sample period is from March 1986 through December 1998.

	Sales			Purchases		
	Quarters 1-3	Quarter 4	t -Test	Quarters 1-3	Quarter 4	t -Test
Extreme Loser Stocks						
External Managers						
All	1.435 (0.063)	1.567 (0.080)	0.131 (8.213)	1.221 (0.024)	1.256 (0.027)	0.033 (4.362)
Winners	1.262 (0.048)	1.152 (0.067)	-0.110 (-6.838)	1.209 (0.052)	1.269 (0.057)	0.060 (4.133)
Losers	1.714 (0.099)	1.858 (0.094)	0.144 (8.077)	1.233 (0.022)	1.254 (0.042)	0.021 (2.069)
Internal Managers						
All	1.242 (0.050)	1.267 (0.059)	0.025 (1.365)	1.250 (0.024)	1.271 (0.032)	0.022 (3.354)
Winners	1.091 (0.089)	1.072 (0.082)	-0.018 (-1.369)	1.321 (0.116)	1.367 (0.139)	0.046 (2.173)
Losers	1.268 (0.095)	1.303 (0.096)	0.033 (2.226)	1.204 (0.100)	1.182 (0.096)	-0.021 (-0.752)
Extreme Winner Stocks						
External Managers						
All	1.363 (0.024)	1.300 (0.027)	-0.062 (-9.080)	1.109 (0.017)	1.104 (0.027)	-0.005 (-0.624)
Winners	1.518 (0.069)	1.563 (0.073)	0.044 (2.199)	1.139 (0.076)	1.141 (0.125)	0.001 (0.066)
Losers	0.951 (0.038)	0.871 (0.053)	-0.080 (-7.661)	0.960 (0.011)	0.959 (0.013)	-0.001 (-0.131)
Internal Managers						
All	1.134 (0.052)	1.103 (0.076)	-0.031 (-2.401)	1.007 (0.026)	0.983 (0.047)	-0.024 (-1.970)
Winners	1.229 (0.076)	1.189 (0.093)	-0.033 (-1.815)	1.027 (0.079)	1.012 (0.082)	-0.012 (-1.039)
Losers	0.983 (0.118)	0.943 (0.084)	-0.046 (-2.476)	0.973 (0.067)	0.939 (0.100)	-0.024 (-1.475)